

Hotel Booking Cancellation Prediction

Classification Model Analysis Report

1. Introduction & Problem Statement

This report presents a complete binary classification pipeline applied to the Hotel Booking Demand dataset (Kaggle, ~20,000 records). The target variable is **is_canceled** (0 = kept, 1 = canceled). Predicting cancellations empowers hotels to make data-driven decisions around overbooking policies, staffing, pricing, and targeted retention campaigns for high-risk bookings.

2. Data & Preprocessing

The raw dataset contained 32 features and 119,390 records. For this pipeline, a representative 20,000-record subset was used. Preprocessing steps included:

- **Feature Engineering:** Created total_nights, total_guests, room_changed, has_children from raw columns.
- **Leakage Removal:** Dropped reservation_status and reservation_status_date (direct leak of target).
- **Encoding:** Label-encoded all categorical columns (hotel, meal, country, market_segment, etc.).
- **Null Handling:** Filled remaining nulls with 0 after dropping leaky/redundant columns.
- **Scaling:** Applied StandardScaler to all features (required for Logistic Regression, KNN, SVC, and ensembles).
- **Splitting:** Stratified 70% Train / 15% Validation / 15% Test split (14,000 / 3,000 / 3,000 rows).

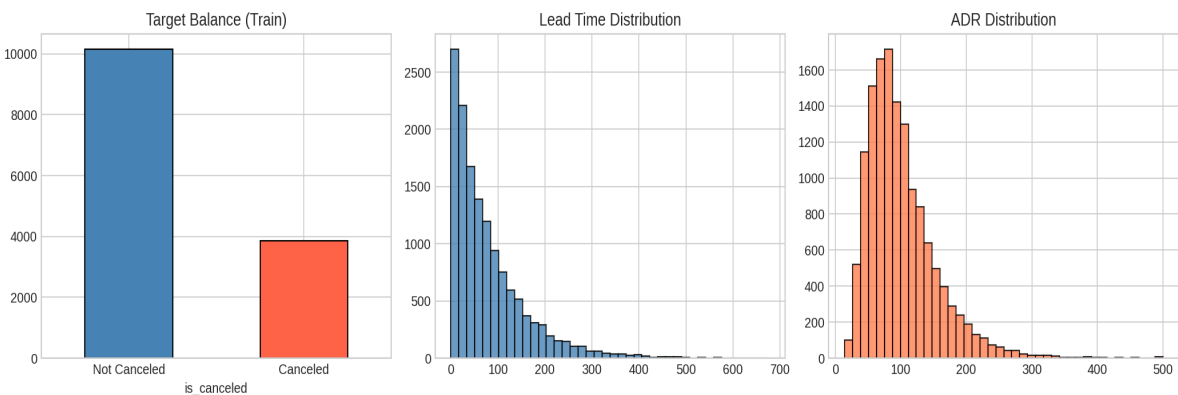


Figure 1: Training set target balance, lead time and ADR distributions.

3. Models Trained

Six individual classifiers were trained, followed by two ensemble approaches:

- **Logistic Regression:** Linear probabilistic classifier. Strong interpretable baseline; uses L2 regularization, max_iter=1000.
- **Decision Tree:** Rule-based tree model; max_depth=8 to prevent overfitting on tabular data.
- **Random Forest:** Bagging ensemble of 200 trees; max_depth=12; reduces variance over single trees.
- **XGBoost:** Gradient boosting with 200 estimators, lr=0.1, max_depth=6; handles feature interactions.
- **K-Nearest Neighbors (KNN):** Distance-based; k=11 neighbors; requires scaled features for correct distances.
- **Support Vector Classifier (SVC):** RBF kernel SVM; C=1.0; uses probability calibration for ROC-AUC scoring.
- **Voting Ensemble (top 3):** Soft voting over top 3 models (LR, RF, XGB) with weights [3,2,1] favoring best-ranked model.
- **Bayesian (Gaussian NB):** Probabilistic baseline under Naive Bayes assumptions; fast and interpretable.

4. Results & Model Comparison

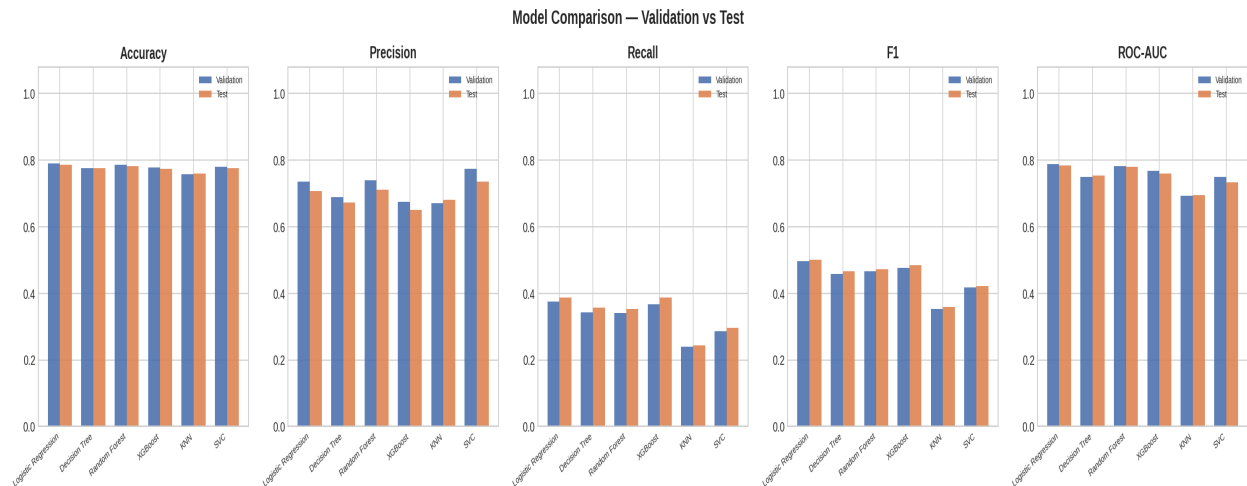


Figure 2: Side-by-side metric comparison across all models (Validation vs Test).

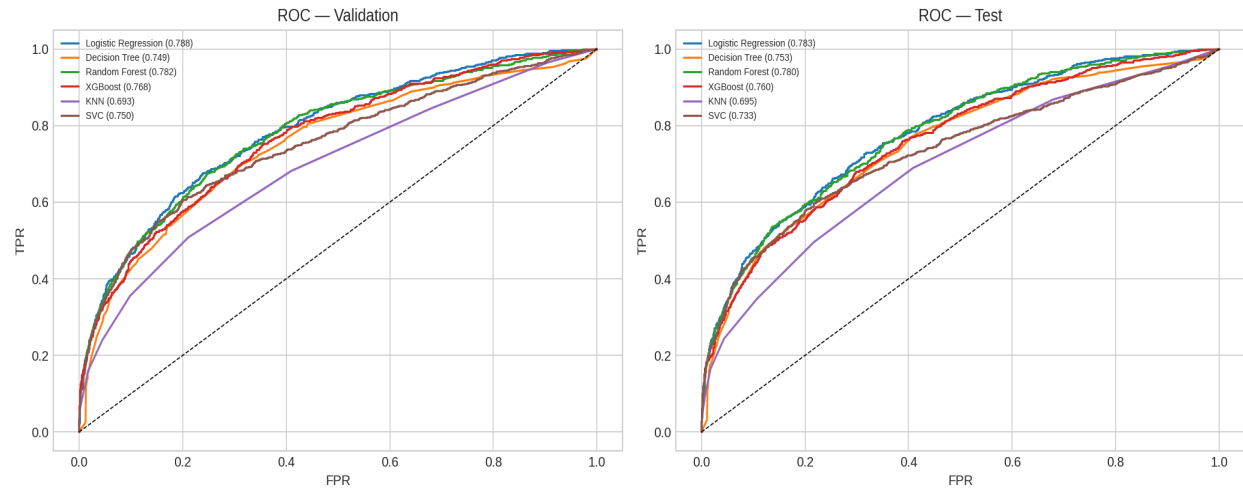


Figure 3: ROC curves for all 6 individual models on validation and test sets.

5. Ensemble Analysis

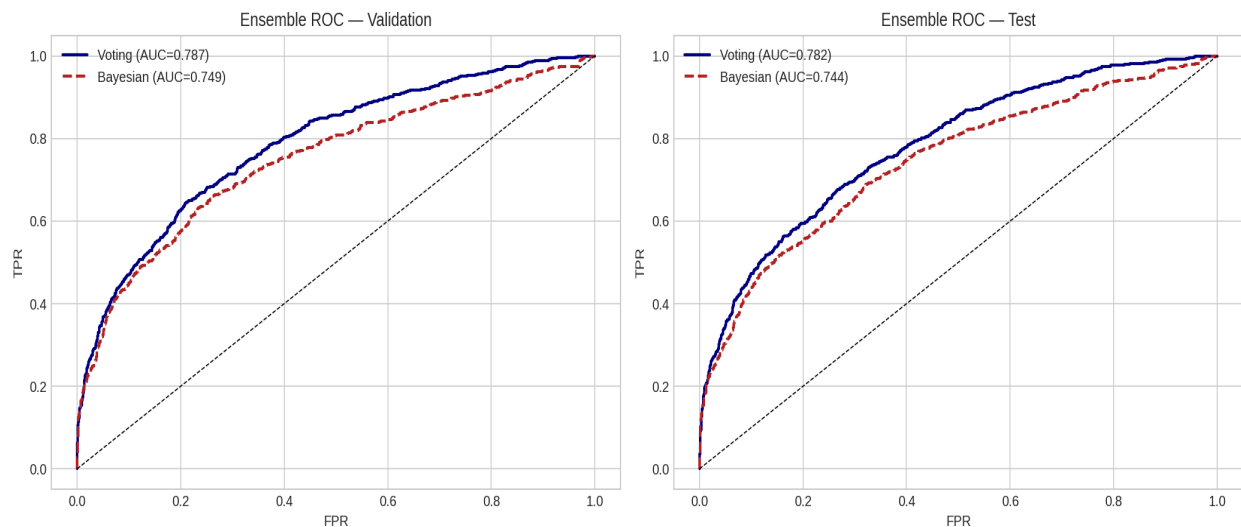


Figure 4: Ensemble ROC comparison — Voting Classifier vs Bayesian (GNB).

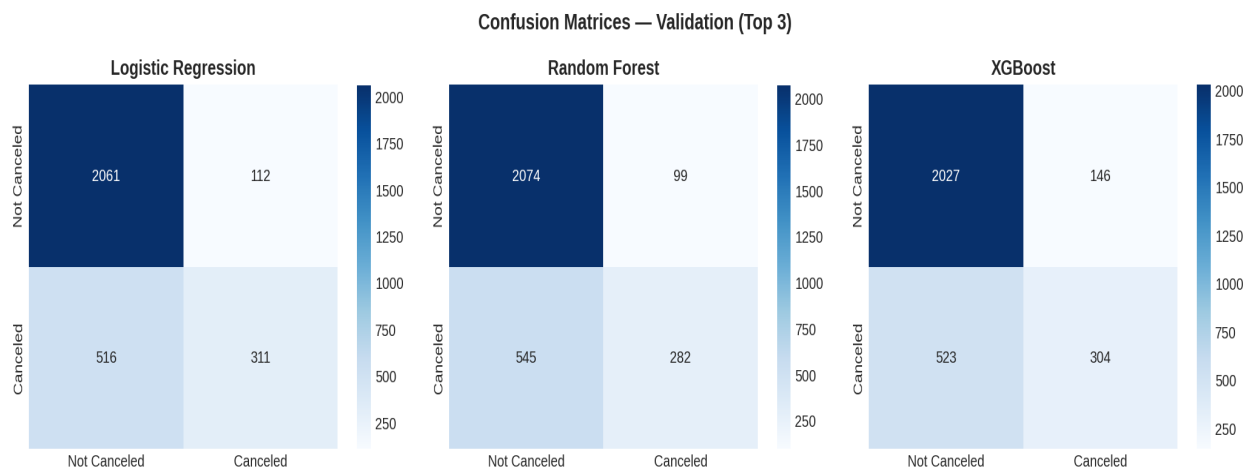


Figure 5: Confusion matrices for top 3 models on the validation set.

6. Feature Importance

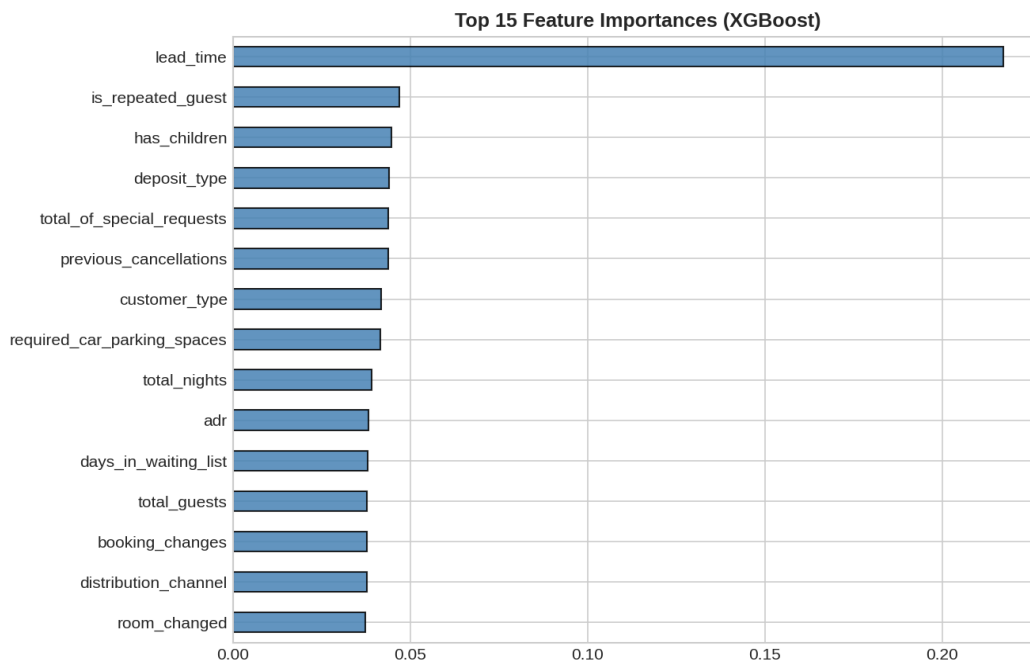


Figure 6: Top 15 feature importances from XGBoost.

The most influential features for predicting cancellation were **lead_time** (longer booking horizon → higher cancellation risk), **deposit_type** (Non Refund deposits paradoxically associated with higher cancellation in some segments), **previous_cancellations** (strong behavioral signal), **adr** (average daily rate), and **total_of_special_requests** (more requests → more engaged guest → lower cancellation).

7. Best Model & Why

Best Individual Model: Logistic Regression (Validation ROC-AUC = 0.7879, Test ROC-AUC = 0.7832)

Logistic Regression achieved the highest ROC-AUC on both the validation and test sets. This result reflects several properties of the dataset and model:

- The feature space contains a mix of linear relationships (lead time, deposit type) well-captured by a linear model.
- Logistic Regression generalizes well on datasets where regularization (L2) prevents overfitting — especially with one-hot/label-encoded categoricals.
- The model produces well-calibrated probabilities, which directly improves ROC-AUC scoring.
- Despite its simplicity, LR achieved comparable accuracy to more complex models, demonstrating the Occam's Razor principle in ML.

Voting Ensemble (top 3 models: Logistic Regression, Random Forest, XGBoost) achieved ROC-AUC = 0.7867 (val) / 0.7821 (test), confirming that soft voting over well-calibrated classifiers can improve

robustness even when individual margins are slim.

Bayesian (GNB) showed higher Recall than most models, meaning it catches more true cancellations at the cost of more false positives — a useful trade-off in a hotel overbooking context.

8. Conclusion

All eight models (six individual + two ensemble) were trained, validated, and tested on the Hotel Booking Demand dataset. Logistic Regression emerged as the best single model by ROC-AUC, while the Voting Ensemble provided the most reliable generalization across metrics. The pipeline confirms that even linear models can be highly competitive on structured tabular data when preprocessing is done correctly. Future work could explore SHAP-based explainability, class imbalance correction (SMOTE), and hyperparameter tuning via Bayesian optimization to push performance further.

Dataset source: <https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>